

Adapting the HeTerogeneity Average (HTA) Index for Numerical Clinical Data

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Overview

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Why heterogeneity matters

- Clinical data often contain substantial variation across subgroups.
- Such variation can influence model stability, fairness, and data needs.
- A quantitative measure is needed to summarize heterogeneity before model training, ideally in an interpretable bounded scale from 0 to 1.

Numerical HTA framework

- 1 Define regions using an Age $\times$ BMI grid.
- 2 Define each trait as a joint combination of categorical variables such as Gender, Ethnicity, EducationLevel, and Smoking.
- 3 Compute local heterogeneity within each region using

$$\text{HTI}_r = - \sum_{c=1}^C p_{rc} \log_C(p_{rc})$$

- 4 Aggregate local scores into a global heterogeneity measure using

$$\text{HTA} = \sum_{r=1}^R \frac{n_r}{n} \text{HTI}_r$$

Trait composition within Age $\times$ BMI regions

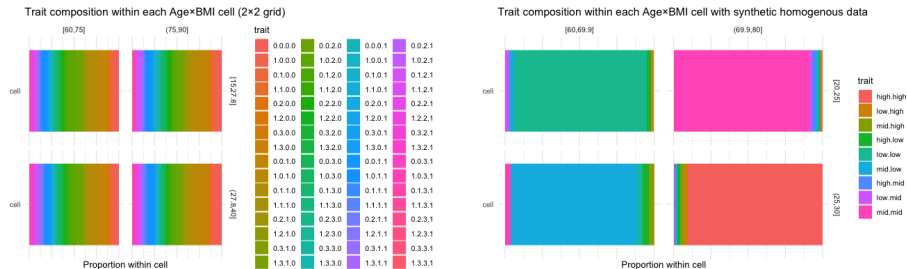


Figure: Within-cell trait composition under two settings. Left: heterogeneous Alzheimer's data, with traits defined by the joint combination of Gender, Ethnicity, EducationLevel, and Smoking. Right: homogeneous synthetic data, with traits defined by joint SBP and LDL categories. The heterogeneous dataset shows more mixed composition across Age $\times$ BMI cells, whereas the homogeneous dataset shows one dominant trait in each cell.

Extension 1: Comparison with alpha Simpson

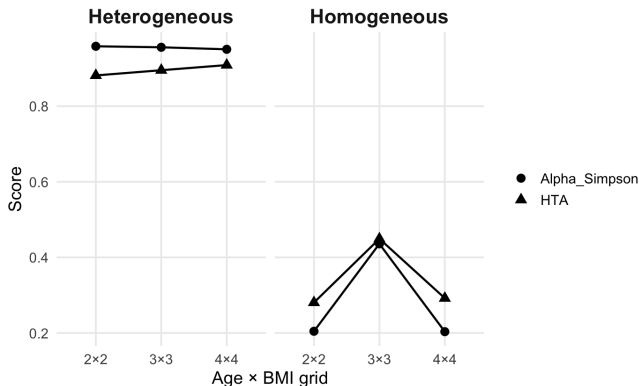


Figure: Comparison of HTA and alpha Simpson across Age $\times$ BMI grid choices under heterogeneous and homogeneous settings. In both cases, alpha Simpson shows a broadly similar pattern to HTA, suggesting that HTA captures a diversity-related signal also reflected by a standard benchmark.

Extension 2: PCA based HTA using PC1 tertiles

- I used PC1 as a simplified trait representation and split it into three tertiles: Low, Mid, and High.
- The resulting HTA was 0.9658 for the heterogeneous dataset and 0.9999 for the homogeneous dataset.
- This suggests that the single PC tertile approach mainly captures how evenly PC1 based categories are mixed within each region, rather than distinguishing heterogeneous from homogeneous structure in the intended way.

Takeaway: When PC1 is split into tertiles, the resulting trait categories are already close to balanced by construction. Under this setup, Age $\times$ BMI regions do not strongly separate these categories, so HTA mainly reflects within-region mixing rather than the intended heterogeneous structure.

Conclusion and limitations

- This project adapted HTA from spatial data to tabular clinical data by defining local regions in Age $\times$ BMI space and measuring trait composition within each region.
- The main analysis showed that heterogeneous and homogeneous settings can have clearly different within-region trait composition under the same framework.
- Extension analyses suggested that HTA is sensitive to how regions and traits are defined, especially when traits are constructed to be balanced by design.

Limitation: The current framework still depends strongly on binning and trait construction, so some extensions may reflect within-region mixing more than the intended notion of heterogeneity.